MILITARY SPECIFICATION

TEST SET, RADIO AN/ARM-54(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall be capable of generating test signals required for test and alignment of AN/ARR-52 Radio Receivers.

1.2 Classification - The test set shall consist of the following items:

Items	Type Designation	Applicable Paragraph
Test Set, Radio	TS-1507(*)/ARM-54	3.4.1
Case, Test Set	CY-3140(*)/ARM-54	3.4.2
Cover, Test Set	CW-584(*)/ARM-54	3.4.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids forms a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment.
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionics Equipment, General Specification

FSC 6625

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STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts
MS17325	Meter, Time Totalizing, 60 and 400 cycle
MS21993	Cable Assemblies, Power Electrical

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

(2) Nonrepairable subassemblies, as outlined in specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment excluding cables shall be a minimum consistent with good design and shall not exceed 38 pounds.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during normal use.

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3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 200 hours of mean operating time between failures when tested and accepted as outlined under the requirements of 4.4.5.

3.2.4 Cables and Connectors - The equipment shall provide for use of cables and connectors in accordance with Specification MIL-T-21200.

3.2.4.1 Cable Assemblies, Power Electrical - Applicable input power cable and connections shall conform to MS21993.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ±10°C)
Altitude	Normal ground
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 V AC and 27.5 ±0.5 V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment except as modified herein.

3.2.11 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 1 minute under standard conditions and 1 minute at extreme service conditions.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200 for Class 3 equipment. The power required shall be either of the following types and shall not exceed the specified amounts.

(1) AC Power, Single Phase, 115 V, 50 to 70 cps and 380 to 420 cps, 30 VA

(2) DC Power, 28 V, 0.6 ampere

3.2.12.1 Voltage Protection - The equipment shall not be damaged by voltages below the voltages specified herein. Over voltage protection shall be provided for the equipment. Damage to the equipment shall be prevented for steady state overload of 100% and transient overload of 500 volts peak with a minimum duration of one microsecond and a maximum duration of 100 milliseconds.

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3.2.13 Time Totalizing Meter - A time totalizing meter conforming as applicable with Specification MIL-M-7793 and MS17325 shall be included in the test set.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified.

3.3.2.1 RF Sweep Frequency - The equipment shall be capable of generating an RF sweep with a center frequency of 168 mc. The sweep width shall be adjustable to a maximum of 30 mc. The maximum sweep output shall be at least 100 mv. It shall be variable to at least -20 db. Two fixed attenuators (20 db, 40 db) shall be provided to extend the variable range to -60 db. Provision shall be made for selecting a frequency marker indicating the center frequency of any one of 16 channels. The detected sweep response over the 30 mc bandwidth shall be ±0.5 db.

3.3.2.2 IF Sweep Frequency - The equipment shall be capable of generating an IF sweep with any one of three center frequencies, 5 mc, 26 mc, and 30 mc. The sweep width shall be a minimum of 0.4 mc at 5 mc, 1.7 mc at 26 mc, and 1.7 mc at 30 mc. The maximum sweep output shall be at least 100 mv. It shall be variable to at least -20 db. Two fixed attenuators (20 db, 40 db) shall be provided to extend the variable range to -60 db. Provision shall be made for the addition of one other frequency. Markers indicating the sweep center frequency shall be provided. The detected sweep response shall be ±0.5 db over the specified bandwidth.

3.3.2.3 IF Oscillator - The equipment shall provide an IF output at any one of nine frequencies:

4.9 mc, 5.0 mc, 5.1 mc - tolerance ±0.5 kc
25.72 mc, 26.00 mc, 26.28 mc - tolerance ±1.0 kc
29.85 mc, 30.00 mc, 30.15 mc - tolerance ±1.0 kc

The maximum output shall be at least 100 mv. It shall be variable to at least -20 db. Two fixed attenuators (20 db, 40 db) shall be provided to extend the variable range to -60 db. Provision for three additional frequencies shall be made.

3.3.2.4 Audio Generator - The equipment shall be capable of generating any one of the following selected frequencies: 10 cps, 100 cps, 200 cps, 800 cps, 2.5 kc, 3 kc, 5 kc, 12 kc, 30 kc, 36 kc, 36 kc, and 50 kc. The frequency tolerance shall be ±3%. When operating into a 300 ohm resistive load, the output level can be adjusted from a minimum of 10 mv rms to a maximum of 2 volts rms.

3.3.3 Attenuated RF Output Characteristics - The equipment shall be capable of amplifying and providing a controlled output for frequencies in the 162 mc to 174 mc range. The RF input voltage shall be 10 mv, +1.5 db, -3.5 db.

Calibrated Output: 10 mv (0 db) to 0.1 uv (100 db) ±2 db continuously variable

Maximum Uncalibrated Output: 100 mv rms minimum

3.3.4 Output Impedance - All external outputs shall work into a nominal 50 ohm impedance unless otherwise specified.

3.4 Detail Requirements

3.4.1 Test Set, Radio TS-1507(*)/ARM-54 - The test set shall meet the following requirements:

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3.4.1.1 Function - This unit shall be capable of generating RF and IF sweep voltages, IF frequencies and audio frequencies required for receiver alignment. This unit, when operated with the AN/ARM-53, shall provide a calibrated RF output voltage for receiver testing and sensitivity measurements.

3.4.1.2 Form Factor - The test set shall be enclosed in an instrument case and the maximum dimensions shall be 14-3/4 inches wide by 8-1/4 inches high by 9-3/4 inches long.

3.4.1.3 Weight - The weight of the test set shall not exceed 23 pounds.

3.4.1.4 Contents - The test set shall consist of the following assemblies:

(1) Calibrated Attenuator

(2) RF Assembly

(3) RF Sweep Generator

(4) IF Sweep Generator

(5) IF Oscillator

(6) Audio Generator

(7) Meter Assembly

(8) Power Supply

3.4.1.4.1 Calibrated Attenuator - There shall be a calibrated RF attenuator controlling the RF output over a 10 millivolt to 0.1 microvolt range.

(1) The attenuator shall operate over a 162 to 174 mc range.

(2) The attenuator shall be calibrated to give a power output accurate to ± 2 db.

3.4.1.4.2 RF Assembly - This assembly shall contain an RF amplifier and power monitoring circuit.

(1) The RF amplifier shall amplify the RF output of the AN/ARM-53 over a 162 to 174 mc range. The calibrated output shall be 10 mv (0 db) to 0.1 uv (100 db) ± 2 db continuously variable.

(2) The power shall be monitored to establish a 10 millivolt reference level to the input of the calibrated attenuator. The maximum output voltage shall be at least 100 mv rms.

3.4.1.4.3 RF Sweep Generator - This assembly shall provide an RF sweep frequency voltage The center frequency of the sweep shall be 168 mc and the maximum sweep width shall be 30 mc. Frequency is swept from 153 mc to 183 mc constant amplitude to within ± 0.5 db.

(1) Marker frequencies shall be provided by an external source.

(2) The output voltage shall be 0.1 volt minimum.

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3.4.1.4.4 IF Sweep Generator - This assembly shall be capable of generating an IF sweep frequency about either of three center frequencies:

5 mc - bandwidth 0.4 mc maximum
26 mc - bandwidth 1.7 mc maximum
30 mc - bandwidth 1.7 mc maximum

The maximum sweep output shall be at least 100 mv. Provision for one additional sweep frequency shall be provided.

3.4.1.4.5 IF Oscillator - This assembly shall be capable of oscillating at any one of nine frequencies:

4.9 mc, 5.0 mc, 5.1 mc - tolerance ±0.5 kc
25.72 mc, 26.00 mc, 26.28 mc - tolerance ±1.0 kc
29.85 mc, 30.00 mc, 30.15 mc - tolerance ±1.0 kc

The maximum output shall be at least 100 mv. Provision for three additional frequencies shall be made.

3.4.1.4.6 Audio Generator - This assembly shall contain a variable frequency audio oscillator and a sweep voltage amplifier.

(1) The audio oscillator shall be capable of oscillation at any one of 12 selected frequencies specified in 3.3.2.4 and having the specified tolerances.

(2) The audio oscillator output shall be variable to a maximum of 2 volts rms across a 300 ohm load.

(3) In the 200 cps position the audio unit shall provide an additional amplification to a nominal 5.3 volts rms for driving the frequency determining circuits of the IF and RF sweep generators. Also, a 0° to 90° phase-adjustable output shall be provided for use with an oscilloscope to display the sweep output.

3.4.1.4.7 Meter Assembly - This assembly shall contain voltage divider circuits and a meter. The deviation sensitivity of the AN/ARM-53 Test Set is ±37.5 kc per sec per volt rms.

(1) The switch shall have seven positions which will determine the meter connection.

(2) The meter shall monitor either the output of the audio detector, RF power level or power supply voltages.

(3) The measured audio detector output shall provide a measure of frequency deviation. The meter scale is calibrated in volts rms with lines to indicate the required deviation levels -22.5 kc per sec., 30 kc per sec. and 75 kc per sec.

3.4.1.4.8 Power Supply - This unit shall contain a power transformer, filter, rectifier, and voltage regulator. The unit shall operate from input voltages of either 115 V AC or 28 V DC.

3.4.1.5 Controls - The following controls shall be on the front panel with identification as underlined below:

Control	Function
METER FNC	Switches meter to monitor either audio level, RF level deviation, or power supply voltages. Controls input to meter and required circuitry to produce reasonable deflection. The meter movement is 100 micro-amperes DC with main scale calibration of 0-2.5.

Step 1 - 28 V internal power supply level. 28 V fed via 499k ohm resistor to produce scale reading times 20.

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Control	Function
Step 2 - 18 V internal power supply level.	18 V fed via 249k ohm resistor to produce scale reading times 10.
Step 3 - 12 V internal power supply level.	12 V fed via 249k ohm resistor to produce scale reading times 10.
Step 4 - OFF	
Step 5 - AUD LEVEL -	Audio output fed to rectifier circuit to produce a reading proportional to rms voltage - Scale reads directly.
Step 6 - RF LEVEL -	A dc current generated internally in the RF amplifier is used merely as a reference to set RF Set control to ensure 10 mv rms into calibrated attenuator.
Step 7 - DEV -	Essentially the same as Audio Level above. Knowledge of the rms audio level implies a particular deviation of the RF output of the AN/ARM-53 Test Set (37.5 kc per sec. per volt rms)
PWR SEL	Selects 28V, OFF or 115V operation
AUDIO FREQ	Selects one of 11 audio frequencies
AUDIO AMP	Varies output level
RF ATTEN	Determines RF output level
RF SET	Calibrates 0 db point of attenuator to 10 MV output
IF Sweep Frequency (MC)	Selects one of three frequencies (5, 26 or 30)
IF Sweep Amplitude (AMP)	Varies output level
IF Oscillator Frequency (FREQ MC)	Selects one of nine frequencies
IF Oscillator Amplitude (AMP)	Varies output level
RF Sweep Amplitude (AMP)	Varies output level
RF Sweep WIDTH	Varies sweep width
Generator Function (GEN FNC)	Selects generator output
Horizontal Sweep PHASE	Varies phase of 200 cps to oscilloscope

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	* Bendix PT01E-8-4S	TS-1507 Power
J2	* Bendix PT01E-3-3P	Power In 28 VDC
J3	* Bendix PT01E-8-3PW	Power In 115 VAC
J4	UG-910A/U	RF Input
J5	UG-910A/U	RF Output
J6	UG-910A/U	IF Sweep Output

* NOTE: Or an approved equivalent.

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Reference Designation	Receptacle Type	Function
J7	UG-910A/U	RF Sweep Output
J8	UG-910A/U	IF Oscillator Output
J9	UG-910A/U	Horizontal Sweep
J10	UG-910A/U	Audio Output

3.4.2 Case, Test Set CY-3140(*)/ARM-54 - The test set shall be provided with a transit case in accordance with Specification MIL-T-21200.

3.4.2.1 Function - The transit case shall provide protection for the contained test equipment.

3.4.2.2 Form Factor - The maximum dimensions of the transit case shall be 9-1/2 inches wide by 11-1/2 inches high by 19 inches long.

3.4.2.3 Weight - The weight of the transit case shall not exceed 9 pounds.

3.4.3 Cover, Test Set CW-584/ARM-54 - The test set shall be furnished with a transit case cover in accordance with Specification MIL-T-21200.

3.4.3.1 Function - The cover shall be used with the transit case to enclose and provide protection for the test equipment.

3.4.3.2 Form Factor - The maximum dimensions of the cover shall be 3-1/2 inches wide by 11-1/2 inches high by 19 inches long.

3.4.3.3 Weight - The weight of the cover shall not exceed 6 pounds.

3.5 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

Test Set, Radio AN/ARM-53

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, including the Sand and Dust Test and modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.

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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests -When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

(4) Reliability Quality Assurance Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests of 4.4.5 by the Government Inspector in accordance with the following: (When reliability tests are required the sampling test shall be deleted.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a *minimum*, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (*This* will be determined by the procuring activity.)

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4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4.5 Reliability Quality Assurance Tests - Unless otherwise stipulated in the contract equipments which have successfully passed the individual tests shall be tested for reliability quality assurance in accordance with Specification MIL-R-23094.

4.4.5.1 Test Level - The Test Level shall be II (from Table I of Specification MIL-R-23094). During each cycle the equipment shall be operated at the following time(s):

4.4.5.2 ACCEPT-Reject Criteria - Figure 12 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.5.3 Detailed Reliability Test Procedure - The contractor shall prepare the detailed reliability Test Procedure and submit it to the procuring activity for approval, the procedure shall include, but not be limited to, the following:

(a) Accept-Reject Criteria

(b) Details of the test cycle

(c) Performance characteristics

(d) Failure Criteria

(e) Preventive Maintenance to be allowed

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4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Use - The equipment shall be used in shop testing of AN/ARR-52 and similar receivers.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1.0 V AC and 27.5 ± 5 V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.5 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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